KAIZEN MATHS

Lesson starter

Remainder and Factor Theorem

Factor Theorem and Solving

Name _____

S1

Is $(x-3)$ a factor of $P(x)=x^3+2x^2-11x-12$?

S2

Given $(x+2)$ is a factor of $P(x)=x^3-3x^2-4x+12$, solve $P(x)=0$.

S3

P(x)= $x^3+kx^2+-2x+56$ and $(x+4)$ is a factor. Find k.

S4

For $P(x)=x^2+px+q$, remainders for $x+4$ and $x+2$ are 36 and 14. Find p, q.

Show your working

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KAIZEN MATHS

Exit ticket

Remainder and Factor Theorem

Factor Theorem and Solving

Name _____

E1

For $P(x) = x^3 + px^2 + qx + -8,\ P(5) = 67$ and $P(7) = 223$. Find $P(x)$.

E2

Is $(x - 3)$ a factor of $P(x) = x^3 - 2x^2 - 5x + 6$?

E3

Given $(x + 3)$ is a factor of $P(x) = x^3 - 13x - 12,$ solve $P(x) = 0$.

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Remainder and Factor Theorem

Factor Theorem and Solving

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Remainder and Factor Theorem

Factor Theorem and Solving

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Teacher answers & working

Remainder and Factor Theorem · Factor Theorem and Solving

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S1

Is $(x-3)$ a factor of $P(x)=x^3+2x^2-11x-12$?

Answer:

$P(3)=0,\ (x-3)$ **is a factor.**

Working:

Step 1: Use the factor theorem. A binomial is a factor exactly when the remainder is 0.

Step 2: The factor (x - 3) gives the root $x=3$.

Step 3: Evaluate $P(3)$.

$P(3)=(3)^3+2(3)^2-11(3)-12$

$P(3)=0$

Step 4: Since $P(3)=0$, the remainder is 0.

Therefore (x - 3) is a factor.

S2

Given $(x+2)$ is a factor of $P(x)=x^3-3x^2-4x+12,$ solve $P(x)=0.$

Answer:

$x=-2,\ 2,\ 3$

Working:

Step 1: Use the known factor (x + 2) to divide the polynomial.

-2	1	-3	-4	12
		-2	10	-12
	1	-5	6	0

Step 2: The bottom row gives the quadratic quotient.

Step 3: Factorise the quotient fully.

$P(x)=(x+2)(x-2)(x-3)$

Step 4: Set each factor equal to zero.

$\boxed{x=-2,\ 2,\ 3}$

S3

P(x)= $x^3+kx^2+-2x+56$ and $(x+4)$ is a factor. Find k.

Answer:

$k=0$

Working:

Since (x + 4) is a factor, $P(-4)=0$.

$-4^3+k(-4)^2+-2(-4)+56=0$

Solving gives $k=0$.

Teacher answers & working

Remainder and Factor Theorem · Factor Theorem and Solving

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S4

For $P(x) = x^2 + px + q$, remainders for $x + 4$ and $x + 2$ are 36 and 14. Find p, q.

Answer:

$p = -5,\ q = 0$

Working:

Use $P(-4) = 36$ and $P(-2) = 14$.

$-4p + q = 20$

$-2p + q = 10$

Solving gives $p = -5$ and $q = 0$.

E1

For $P(x) = x^3 + px^2 + qx + -8,\ P(5) = 67$ and $P(7) = 223$. Find $P(x)$.

Answer:

$P(x) = x^3 + -3x^2 + 5x + -8$

Working:

Use the remainder theorem with $x = 5$ and $x = 7$.

$25p + 5q = -50$

$49p + 7q = -112$

Solving gives $p = -3$, $q = 5$.

E2

Is $(x - 3)$ a factor of $P(x) = x^3 - 2x^2 - 5x + 6$?

Answer:

$P(3) = 0,\ (x - 3)$ is a factor.

Working:

Step 1: Use the factor theorem. A binomial is a factor exactly when the remainder is 0.

Step 2: The factor (x - 3) gives the root $x = 3$.

Step 3: Evaluate $P(3)$.

$P(3) = (3)^3 - 2(3)^2 - 5(3) + 6$

$P(3) = 0$

Step 4: Since $P(3) = 0$, the remainder is 0.

Therefore (x - 3) is a factor.

Teacher answers & working

Remainder and Factor Theorem · Factor Theorem and Solving

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E3

Given $(x+3)$ is a factor of $P(x)=x^3-13x-12,$ solve $P(x)=0.$

Answer:

$x=-3,\ 4,\ -1$

Working:

Step 1: Use the known factor (x + 3) to divide the polynomial.

-3	1	0	-13	-12
		-3	9	12
	1	-3	-4	0

Step 2: The bottom row gives the quadratic quotient.

Step 3: Factorise the quotient fully.

$P(x)=(x+3)(x-4)(x+1)$

Step 4: Set each factor equal to zero.

$\boxed{x=-3,\ 4,\ -1}$